

Example 9.20

An E-MOSFET amplifier circuit has the following data.

$$\begin{aligned} V_T &= 4 \text{ V} & I_{D(\text{on})} &= 4 \text{ mA} \\ V_{GS(\text{on})} &= 7 \text{ V} & y_{os} &= 25 \text{ } \mu\text{S}. \end{aligned}$$

Calculate the values of k , g_m and r_d .

Solution

$$\begin{aligned} I_D &= k [V_{GS} - V_{GS(\text{Th})}]^2 \\ \Rightarrow I_{D(\text{on})} &= k [V_{GS(\text{on})} - V_{GS(\text{Th})}]^2 \end{aligned}$$

$$k = \frac{I_{D(\text{on})}}{[V_{GS(\text{on})} - V_{GS(\text{Th})}]^2}$$

Note that

$$V_T = V_{GS(\text{Th})}$$

$$\begin{aligned} k &= \frac{4 \text{ mA}}{[7 \text{ V} - 4 \text{ V}]^2} \\ &= 0.44 \times 10^{-3} \text{ A/V}^2 \end{aligned}$$

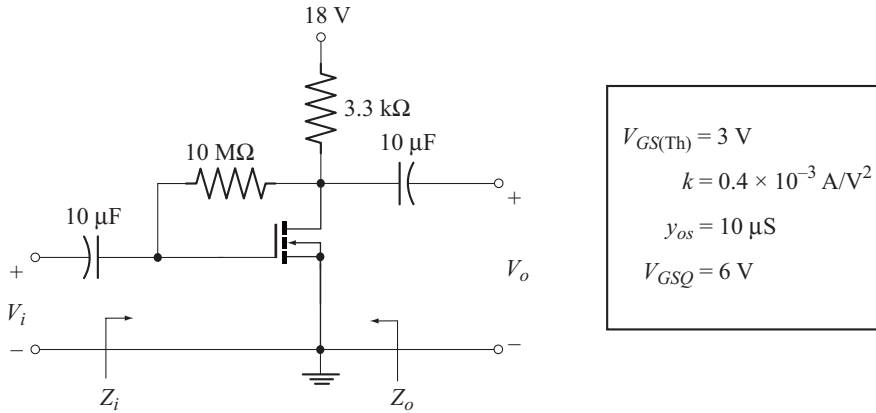
$$\begin{aligned} g_m &= 2k [V_{GSQ} - V_{GS(\text{Th})}] \\ V_{GSQ} &= V_{GS(\text{on})} \\ g_m &= 2 [0.44 \times 10^{-3} \text{ A/V}^2] [7 \text{ V} - 4 \text{ V}] \\ &= 2.64 \text{ mS}. \end{aligned}$$

$$\begin{aligned} r_d &= \frac{1}{y_{os}} \\ &= \frac{1}{25 \text{ } \mu\text{S}} = 40 \text{ k}\Omega \end{aligned}$$

Example 9.21

For the E-MOSFET drain feed back amplifier shown below:

- Calculate g_m and r_d .
- Determine Z_i and Z_o .
- Calculate A_v .
- Find V_o if $V_i = 5 \text{ mV}$.
- Calculate Z_i , Z_o , A_v and V_o , neglecting the effect of r_d and compare the results.

**Solution**

$$\begin{aligned} (a) \quad g_m &= 2 \text{ k} [V_{GSQ} - V_{GS(Th)}] \\ &= (2) (0.4 \times 10^{-3} \text{ A/V}^2) [6 \text{ V} - 3 \text{ V}] = 2.4 \text{ mS}. \end{aligned}$$

$$r_d = \frac{1}{y_{os}} = \frac{1}{10 \text{ mS}} = 100 \text{ k}\Omega$$

$$\begin{aligned} (b) \quad Z_i &= \frac{R_F + [r_d \parallel R_D]}{1 + g_m [r_d \parallel R_D]} \\ &= \frac{10 \text{ M}\Omega + [100 \text{ k}\Omega \parallel 3.3 \text{ k}\Omega]}{1 + [2.4 \text{ mS}] [100 \text{ k}\Omega \parallel 3.3 \text{ k}\Omega]} = 1.15 \text{ M}\Omega \end{aligned}$$

$$\begin{aligned} Z_o &= R_F \parallel r_d \parallel R_D \\ &= 10 \text{ M}\Omega \parallel 100 \text{ k}\Omega \parallel 3.3 \text{ k}\Omega = 3.29 \text{ k}\Omega \end{aligned}$$

$$\begin{aligned} (c) \quad A_v &= \frac{\hat{e}}{\hat{e}} \frac{1}{R_F} - g_m \hat{u} [R_F \parallel r_d \parallel R_D] \\ &= \frac{\hat{e}}{\hat{e}} \frac{1}{10 \text{ M}\Omega} - 2.4 \text{ mS} \hat{u} [10 \text{ M}\Omega \parallel 100 \text{ k}\Omega \parallel 3.3 \text{ k}\Omega] \\ &= -7.89 \end{aligned}$$

$$\begin{aligned} (d) \quad V_o &= A_v V_i \\ &= [-7.89] [5 \text{ mV}] = -39.45 \text{ mV} \end{aligned}$$

(e) *When the effect of r_d is neglected*

$$R_F = 10 \text{ M}\Omega \quad \text{and} \quad r_d \parallel R_D = 3.19 \text{ k}\Omega$$

Note that,

$$R_F \gg r_d \parallel R_D$$

$$r_d = 100 \text{ k}\Omega \quad \text{and} \quad 10 R_D = 33 \text{ k}\Omega$$

Also note that $r_d > 10 R_D$.

Hence we can use approximate results by neglecting r_d .

$$\begin{aligned}
 Z_i &\approx \frac{R_F}{1 + g_m R_D} \\
 &= \frac{10 \text{ MW}}{1 + [2.4 \text{ mS}] [3.3 \text{ kW}]} = 1.12 \text{ M}\Omega \\
 Z_o &\approx R_D = 3.3 \text{ k}\Omega \\
 A_v &\approx -g_m R_D = -[2.4 \text{ mS}] [3.3 \text{ k}\Omega] = -7.92 \\
 V_o &= A_v V_i = [-7.92] [5 \text{ mV}] = -39.6 \text{ mV}.
 \end{aligned}$$

The results are compared in the following table.

Parameter	With r_d	With out r_d
Z_i	1.15 M Ω	1.12 M Ω
Z_o	3.29 M Ω	3.3 M Ω
A_v	-7.89	-7.92

Note that the results are very close since the conditions $R_F \gg r_d \parallel R_D$ and $r_d > 10 R_D$ are satisfied.

◆ 9.21 E-MOSFET COMMON-SOURCE AMPLIFIER USING VOLTAGE DIVIDER CONFIGURATION

Figure 9.41 shows the circuit of E-MOSFET common-source amplifier using voltage divider bias.

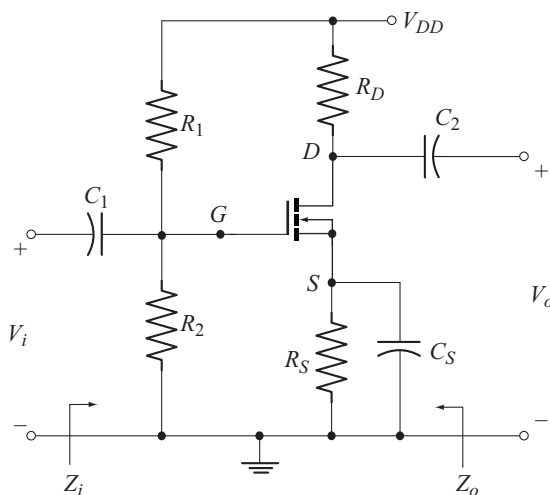


Fig. 9.41 E-MOSFET voltage divider configuration

The ac equivalent circuit is drawn in Fig. 9.42 by

- reducing V_{DD} to zero
- replacing the capacitors C_1 , C_2 and C_S with their short circuit equivalents and
- replacing E-MOSFET with its small-signal ac model

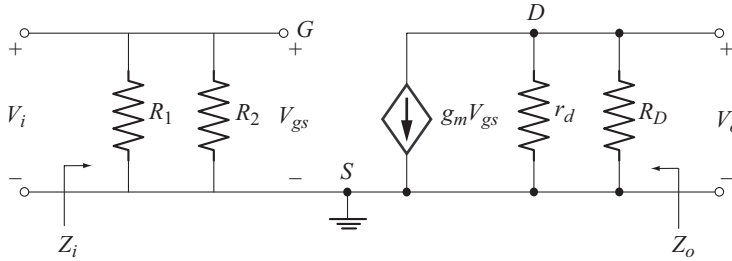


Fig. 9.42 AC equivalent circuit of E-MOSFET voltage divider configuration

This circuit is exactly identical that of Fig. 9.20 analysed in section 9.12, for JFET voltage divider bias configuration. The results are reproduced below for reference.

When the effect of r_d is included

$$Z_i = R_1 \parallel R_2 \quad (9.118)$$

$$Z_o = r_d \parallel R_D \quad (9.119)$$

$$A_v = -g_m [r_d \parallel R_D] \quad (9.120)$$

The negative sign in A_v reveals that V_o and V_i are 180° out of phase.

When the effect of r_d is neglected [For $r_d \geq 10 R_D$]

$$Z_i = R_1 \parallel R_2 \quad (9.121)$$

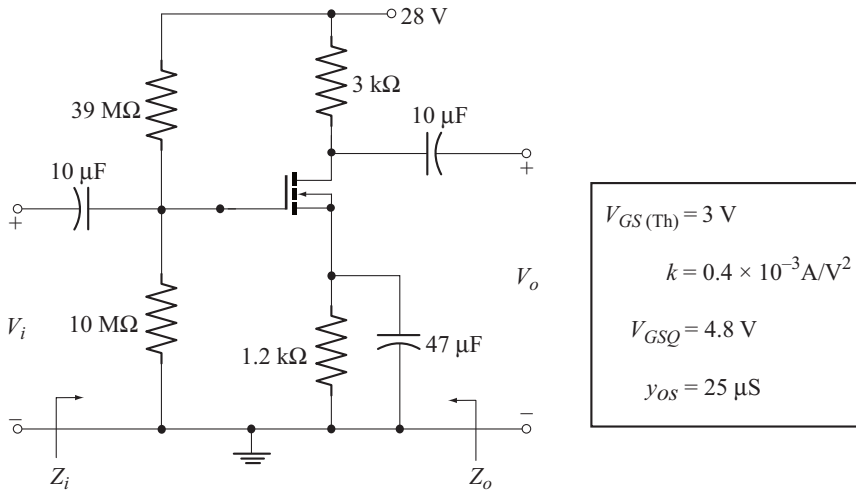
$$Z_o \approx R_D \quad (9.122)$$

$$A_v \approx -g_m R_D \quad (9.123)$$

Example 9.22

For the E-MOSFET voltage divider configuration shown

- Calculate Z_i and Z_o .
- Calculate A_v .
- Find V_o if $V_i = 1$ mV.
- Calculate Z_i , Z_o , A_v and V_o , neglecting the effect of r_d .

**Solution**

$$g_m = 2k [V_{GSQ} - V_{GS(Th)}]$$

$$= 2 [0.4 \times 10^{-3} \text{ A/V}^2] [4.8 \text{ V} - 3 \text{ V}] = 1.44 \text{ mS}$$

$$r_d = \frac{1}{y_{os}} = \frac{1}{25 \mu\text{S}} = 40 \text{ k}\Omega$$

$$(a) \quad Z_i = R_1 \parallel R_2 = 39 \text{ M}\Omega \parallel 10 \text{ M}\Omega = 7.95 \text{ M}\Omega$$

$$Z_o = r_d \parallel R_D = 40 \text{ k}\Omega \parallel 3 \text{ k}\Omega = 2.79 \text{ k}\Omega$$

$$(b) \quad A_v = -g_m [r_d \parallel R_D] = -[1.44 \text{ mS}] [2.79 \text{ k}\Omega] = -4.01$$

$$(c) \quad V_o = A_v V_i = [-4.01] [1 \text{ mV}] = -4.01 \text{ mV}$$

$$(d) \quad r_d = 40 \text{ k}\Omega \quad \text{and} \quad 10 R_D = 30 \text{ k}\Omega$$

Since $r_d > 10 R_D$, the effect of r_d can be neglected.

$$Z_i = R_1 \parallel R_2 = 7.95 \text{ M}\Omega$$

$$Z_o \approx R_D = 3 \text{ k}\Omega$$

$$A_v \approx -g_m R_D = -[1.44 \text{ mS}] [3 \text{ k}\Omega] = -4.32$$

$$V_o = A_v V_i = [-4.32] [1 \text{ mV}] = -4.32 \text{ mV}.$$

◆ 9.22 DESIGN OF FET AMPLIFIERS

Design of FET amplifiers involve the calculation of circuit elements R_D , R_G , R_S , C_1 , C_2 and C_S to meet the required ac and dc specifications.

In most of the cases, drain supply voltage V_{DD} is known. The values of R_D and R_S are calculated from the gain and biasing requirements. In majority of the circuits, R_G decides the input impedance.

The capacitors C_1 , C_2 and C_S are calculated to meet the required frequency response specifications. The following examples illustrate the design of some of the FET amplifier networks.

Example 9.23

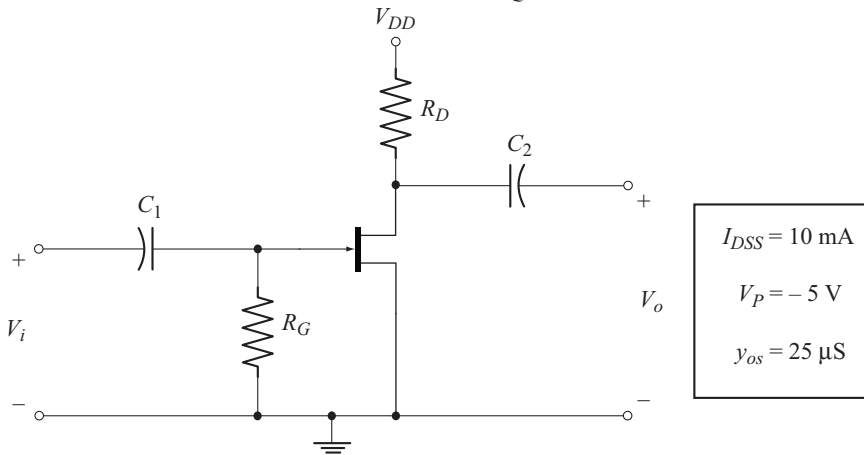
Design the fixed bias FET common-source amplifier shown below to meet the following requirements.

Magnitude of voltage gain, $|A_V| : 12$

Input impedance, $Z_i : 10 \text{ M}\Omega$

Available drain supply voltage, $V_{DD} : 40 \text{ V}$

For the circuit designed, calculate the values of V_{DSQ} and Z_o .



Solution

Selection of R_G

For the given circuit

$$Z_i = R_G$$

Since

$$Z_i = 10 \text{ M}\Omega$$

$$\text{Select } R_G = 10 \text{ M}\Omega$$

Selection of R_D

$$A_V = -g_m [R_D \parallel r_d] \quad (\text{A})$$

$$g_m = g_{m_o} \left[1 - \frac{V_{GSQ}}{V_p} \right]$$

For the given circuit, $V_{GSQ} = 0$ [since $R_S = 0$].

$$\therefore g_m = g_{m_o} = \frac{2I_{DSS}}{|V_p|} = \frac{(2)(10 \text{ mA})}{5 \text{ V}}$$

$$\therefore g_m = 4 \text{ mS.}$$

Given, $A_v = 12$

Since common source configuration is an inverting amplifier, A_v is negative

$$\therefore \quad A_v = -12$$

$$r_d = \frac{1}{y_{os}} = 40 \text{ k}\Omega.$$

From Equation (A)

$$-12 = -4 \text{ mS} [R_D \parallel r_d]$$

$$R_D \parallel r_d = 3 \text{ k}\Omega$$

$$\left[\frac{R_D r_d}{R_D + r_d} \right] = 3 \text{ k}\Omega$$

Solving we get

$$R_D = 3.24 \text{ k}\Omega$$

Output impedance, Z_o

$$Z_o = R_D \parallel r_d = 3.24 \text{ k}\Omega \parallel 40 \text{ k}\Omega = 3 \text{ k}\Omega$$

Calculation of V_{DSQ}

Applying KVL to the drain circuit, we have

$$V_{DD} = I_{DQ} R_D + V_{DSQ} \quad (\text{B})$$

$$I_{DQ} = I_{DSS} \left[1 - \frac{V_{GSQ}}{V_p} \right]$$

$$= I_{DSS} = 10 \text{ mA} \quad [\text{Since } V_{GSQ} = 0]$$

Now from Equation (B), we get

$$V_{DD} = I_{DQ} R_D + V_{DSQ} \quad (\text{B})$$

$$40 \text{ V} = (10 \text{ mA}) (3.24 \text{ k}\Omega) + V_{DSQ}$$

$$V_{DSQ} = 7.6 \text{ V}$$

Example 9.24

Design the self bias FET common-source amplifier shown below to meet the following requirements:

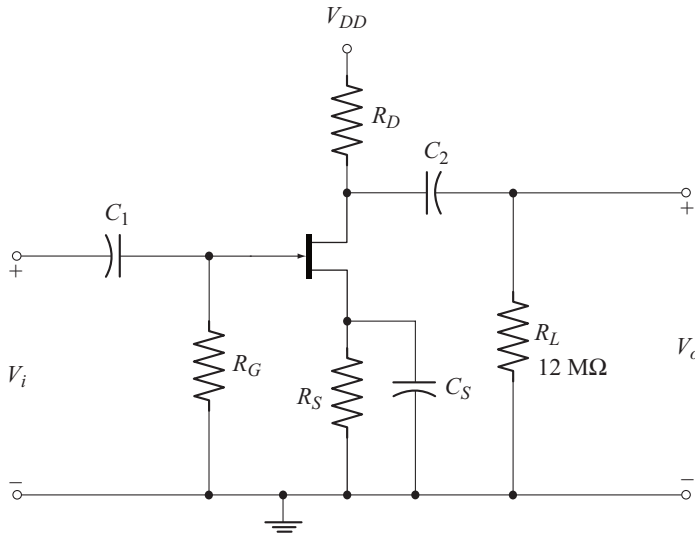
Magnitude of voltage gain, $|A_v| : 10$

Input impedance, $Z_i : 5 \text{ M}\Omega$

Quiescent gate to source voltage, $V_{GSQ} : \frac{V_p}{3}$

Available drain supply voltage, $V_{DD} : 20 \text{ V}$

For the circuit designed calculate the values of V_{DSQ} and Z_o .



$$I_{DSS} = 12 \text{ mA}$$

$$V_P = -3 \text{ V}$$

$$r_d = 40 \text{ k}\Omega$$

Solution

Selection of R_G

For the given circuit,

$$Z_i = R_G$$

But

$$Z_i = 5 \text{ M}\Omega$$

$\therefore$ We select

$$R_G = 5 \text{ M}\Omega$$

Selection of R_D

$$A_v = -g_m [R_D \parallel r_d \parallel R_L]$$

Since $R_L = 12 \text{ M}\Omega$ which is very high relative to r_d , we can treat it as an open circuit.

$$\therefore A_v \approx -g_m [R_D \parallel r_d] \quad (\text{A})$$

$$g_m = g_{m_o} \left[1 - \frac{V_{GS_Q}}{V_P} \right]$$

$$V_{GS_Q} = \frac{V_P}{3} \Rightarrow \frac{V_{GS_Q}}{V_P} = \frac{1}{3}$$

$$g_{m_o} = \frac{2 I_{DSS}}{|V_P|} = \frac{(2)(12 \text{ mA})}{3 \text{ V}} = 8 \text{ mS}$$

$$g_m = 8 \text{ mS} \left[1 - \frac{1}{3} \right] = 5.33 \text{ mS}$$

Now from Equation (A), we have

$$-10 = [-5.33 \text{ mS}] [R_D \parallel r_d]$$

$$R_D \parallel r_d = 1.876 \text{ k}\Omega$$

$$\frac{R_D \cdot r_d}{R_D + r_d} = 1.876 \text{ k}\Omega$$

$$\frac{R_D (40 \text{ k}\Omega)}{R_D + 40 \text{ k}\Omega} = 1.876 \text{ k}\Omega$$

Solving we get, $R_D = 1.968 \text{ k}\Omega$

Calculation of R_S

$$V_{GS_Q} = -I_{D_Q} R_S \quad (\text{B})$$

$$V_{GS_Q} = \frac{V_P}{3} = \frac{-3 \text{ V}}{3} = -1 \text{ V}$$

$$I_{D_Q} = I_{DSS} \left[1 - \frac{V_{GS_Q}}{V_P} \right]^2$$

$$= 12 \text{ mA} \left[1 - \frac{-1 \text{ V}}{-3 \text{ V}} \right]^2 = 6.66 \text{ mA}$$

From Equation (B),

$$R_S = -\frac{V_{GS_Q}}{I_{D_Q}}$$

$$R_S = -\frac{-1 \text{ V}}{6.66 \text{ mA}} = 150 \text{ }\Omega$$

Output impedance, Z_o

$$Z_o = R_D \parallel r_d \parallel R_L$$

$$= 1.968 \text{ k}\Omega \parallel 40 \text{ k}\Omega \parallel 12 \text{ M}\Omega$$

$$\approx 1.968 \text{ k}\Omega \parallel 40 \text{ k}\Omega = 1.875 \text{ k}\Omega.$$

Drain-source voltage, V_{DS_Q}

Applying KVL to the drain circuit, we have

$$V_{DD} = I_{D_Q} R_D + V_{DS_Q} + I_{D_Q} R_S$$

$$20 \text{ V} = (6.66 \text{ mA}) (1.968 \text{ k}\Omega) + V_{DS_Q} + 1 \text{ V}$$

$$V_{DS_Q} = 5.89 \text{ V}$$

Example 9.25

Design the self biased FET common-source amplifier with unbypassed R_S , shown below to meet the following requirements:

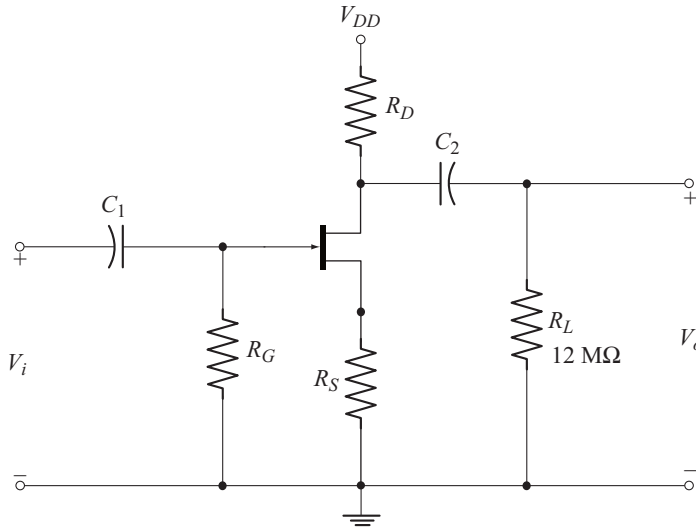
Magnitude of voltage gain, $|A_v| : 8$

Input impedance $Z_i : 2 \text{ M}\Omega$

Quiescent gate to source voltage $V_{GSQ} : \frac{V_P}{4}$

Available drain supply voltage $V_{DD} : 30 \text{ V}$

For the circuit designed, calculate the values of Z_o and V_{DSQ}



$$I_{DSS} = 10 \text{ mA}$$

$$V_P = -5 \text{ V}$$

$$r_d = 50 \text{ k}\Omega$$

Solution

Selection of R_G

For the given circuit,

$$Z_i = R_G$$

But

$$Z_i = 2 \text{ M}\Omega$$

$\therefore$ We select

$$R_G = 2 \text{ M}\Omega$$

Selection of R_S

$$V_{GSQ} = -I_{DQ} R_S \quad (A)$$

$$V_{GSQ} = \frac{V_P}{4} = \frac{-5\text{V}}{4} = -1.25 \text{ V}$$

$$I_{DQ} = I_{DSS} \left[\frac{V_{GSQ}}{V_P} \right]^2$$

$$= 10 \text{ mA} \left[\frac{-1.25 \text{ V}}{-5 \text{ V}} \right]^2 = 5.625 \text{ mA}$$

From Equation (A)

$$R_S = - \frac{V_{GS_Q}}{I_{D_Q}} = - \frac{-1.25\text{V}}{5.625\text{ mA}} = 222.2\ \Omega$$

Selection of R_D

To simplify the calculation, for the moment, let us assume that $r_d \geq 10 [R_D + R_S]$. we will justify our assumption after the calculation of R_D .

$$\text{Now } A_V \approx - \frac{g_m R_D}{1 + g_m R_S} \quad (\text{B})$$

R_L is not considered since it is relatively larger.

$$g_m = g_{m_o} \left[1 - \frac{V_{GS_Q}}{V_p} \right]$$

$$g_{m_o} = \frac{2 I_{DSS}}{|V_p|} = \frac{2 [10\text{ mA}]}{5\text{V}} = 4\text{ mS}$$

$$g_m = 4\text{ mS} \left[1 - \frac{-1.25\text{V}}{-5\text{V}} \right] = 3\text{ mS}.$$

Now from Equation (B), we have

$$\begin{aligned} R_D &= - \frac{A_V [1 + g_m R_S]}{g_m} \\ &= - \frac{[-8] [1 + (3\text{mS})(222.2\text{W})]}{3\text{mS}} = 4.44\text{ k}\Omega \end{aligned}$$

Justification of Assumption

$$\begin{aligned} r_d &= 50\text{ k}\Omega \\ 10 [R_D + R_S] &= 10 [4.44\text{ k}\Omega + 222.2\ \Omega] = 46.62\text{ k}\Omega \end{aligned}$$

Observe that $r_d > 10 [R_D + R_S]$, which justifies our assumption.

Output Impedance

Since $r_d > 10 R_D$

$$Z_o \approx R_D$$

$\therefore$

$$Z_o = 4.44\text{ k}\Omega$$

Drain to Source Voltage

Applying KVL to the drain circuit we have

$$V_{DD} = I_{D_Q} R_D + V_{DS_Q} + I_{D_Q} R_S$$

$$30\text{ V} = (5.625\text{ mA}) (4.44\text{ k}\Omega) + V_{DS_Q} + (5.625\text{ mA}) (222.2\ \Omega)$$

$$V_{DS_Q} = 3.775\text{ V}.$$



Exercise Problems

- 9.1 A common-source JFET amplifier using self-bias with bypassed R_S has $R_G = 1 \text{ M}\Omega$, $R_S = 2.7 \text{ k}\Omega$, $R_D = 3.9 \text{ k}\Omega$, $g_m = 2800 \text{ }\mu\text{S}$ and $r_d = 40 \text{ k}\Omega$. Calculate Z_i , Z_o , A_v and A_i .
- 9.2 Repeat the previous problem with R_S unbypassed.
- 9.3 A JFET source-follower has $R_G = 5 \text{ M}\Omega$, $R_D = 2 \text{ k}\Omega$, $R_S = 2.7 \text{ k}\Omega$, $g_m = 0.8 \text{ mS}$ and $r_d = 40 \text{ k}\Omega$. Calculate Z_i , Z_o , A_v and A_i .
- 9.4 A JFET common-gate amplifier has $R_G = 1.2 \text{ M}\Omega$, $R_D = 3 \text{ k}\Omega$, $g_m = 2 \text{ mS}$ and $r_d = 50 \text{ k}\Omega$. Calculate Z_i , Z_o and A_v .
- 9.5 Design a self-biased common-source JFET amplifier with bypassed R_S which has $|A_v| = 8$ and $Z_i = 10 \text{ M}\Omega$. For the FET, $g_m = 5 \text{ mS}$, $r_d = 50 \text{ k}\Omega$ and $V_{DD} = 22 \text{ V}$.

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